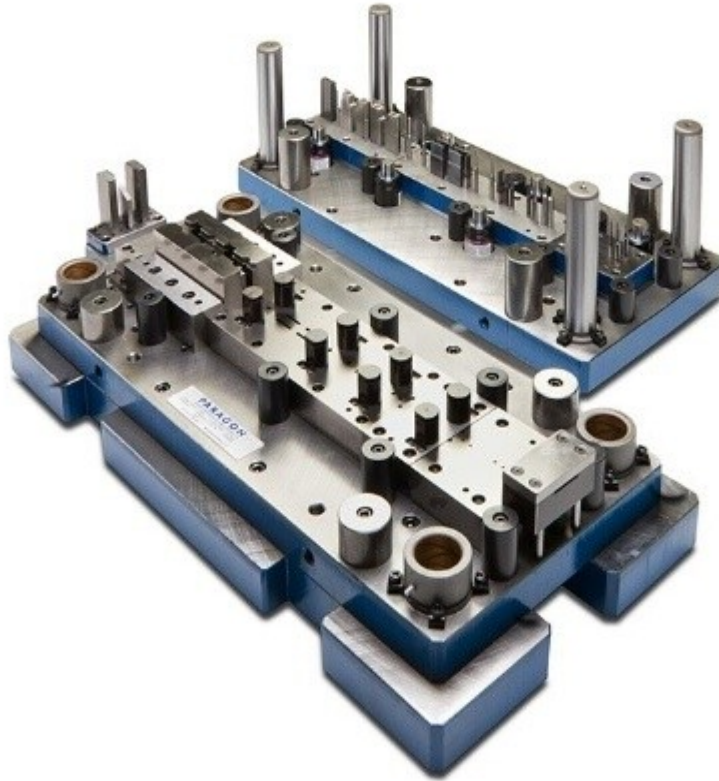


Qualification Pack



Die & Mould Maker

QP Code: TNSD/CSC/Q0103

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

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Qualification Pack

TNSD/CSC/Q0103: Die & Mould Maker

Brief Job Description

A Die & Mould Maker is an expert in process planning, manufacturing, and design of various dies and moulds, including press tools, jigs, fixtures, and die-casting dies. This role requires advanced skills in CNC programming and operation, CAD/CAM software, material analysis, and economic principles.

Personal Attributes

Upon successful completion, learners will be qualified to become experts in the process planning, manufacturing, and design of dies & moulds, jigs & fixtures, press tools, and die casting dies. They will secure employment in engineering, manufacturing, machine tools, automobile, and all types of machinery industries, and also be equipped for entrepreneurship.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [TNSD/CSC/N0128: Manufacture and Try-out Press Tools, Jigs & Fixtures.](#)
2. [TNSD/CSC/N0120: Design and Manufacture Press Tools for Sheet Metal Components](#)
3. [TNSD/CSC/N0121: Design and Utilize Jigs & Fixtures for Mass Production](#)
4. [TNSD/CSC/N0122: Program and Operate Advanced Multi-Axis CNC Machines](#)
5. [TNSD/CSC/N0123: Generate Basic 2D/3D Models and Toolpaths using Master cam](#)
6. [TNSD/CSC/N6912: Create Basic 2D/3D Models using SolidWorks](#)
7. [TNSD/CSC/N0124: Design Complex Press Tools, Jigs & Fixtures using Advanced SolidWorks](#)
8. [TNSD/CSC/N0125: Analyze Material Stress, Strain, and Thermal Effects](#)
9. [TNSD/CSC/N0126: Apply Core Principles of Organizational Management](#)
10. [TNSD/CSC/N0127: Perform Tool & Die Cost Estimation and Economic Analysis](#)
11. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Capital Goods
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Qualification Pack

Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
Country	India
NSQF Level	4.5
Credits	40
Aligned to NCO/ISCO/ISIC Code	3115.1301
Minimum Educational Qualification & Experience	Previous relevant Qualification of NSQF Level (Certificate in Mould Maker – 4)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	1.0
Reference code on NQR	QG-4.5-CG-04459-2025-V1-TNSDC
NQR Version	1.0

Qualification Pack

TNSD/CSC/N0128: Manufacture and Try-out Press Tools, Jigs & Fixtures.

Description

Understand the manufacturing of press tools, jigs & Fixtures by using Conventional, CNC machines and different soft wares. Try-out of press tools, jigs and fixtures.

Scope

The scope covers the following :

- Understand the manufacturing of press tools, jigs & Fixtures by using Conventional, CNC machines and different soft wares

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Learn the standard grades of sheet metal used for processing in Press Tools and general Press Tool operation ,construction and features
- PC2.** Understand the concept of theory of shearing ,cutting force and cutting clearance calculation for different operations ,Strip layout and economy factor
- PC3.**
 - Understand cutting elements- Punch, Die block. Its types and importance.
 - Effect of land and angular clearance. Effect of force in Buckling of punches.
- PC4.** Understand the functions of punch holder, Stripper and pilot and its types.
- PC5.** Learn to Select the right type of gauges, stoppers and fasteners
- PC6.** Understand the usage of side cutter and location of shank in a press tool
- PC7.** Perform Design and drawing of open die and guide plate press tools .
- PC8.** Demonstrate Design and drawing of progressive type of press tools with cutting and Non-cutting operations.
- PC9.** Perform Design and drawing of compound and combination press tool.
- PC10.** Demonstrate design and drawing of fine blanking tool

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the concept of theory of shearing ,cutting force and cutting clearance calculation for different operations ,Strip layout and economy factor
- KU2.**
 - Understand cutting elements- Punch, Die block. Its types and importance.
 - Effect of land and angular clearance. Effect of force in Buckling of punches.
- KU3.** Understand the functions of punch holder, Stripper and pilot and its types.

Generic Skills (GS)

User/individual on the job needs to know how to:

Logo
Not
Available



Qualification Pack

GS1. Learn to
Select
the right
type of
gauges,
stoppers
and
fasteners.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	100	-	-
PC1. Learn the standard grades of sheet metal used for processing in Press Tools and general Press Tool operation ,construction and features	15	-	-	-
PC2. Understand the concept of theory of shearing ,cutting force and cutting clearance calculation for different operations ,Strip layout and economy factor	20	-	-	-
PC3. • Understand cutting elements- Punch, Die block. Its types and importance. • Effect of land and angular clearance. Effect of force in Buckling of punches.	20	-	-	-
PC4. Understand the functions of punch holder, Stripper and pilot and its types.	20	-	-	-
PC5. Learn to Select the right type of gauges, stoppers and fasteners	10	-	-	-
PC6. Understand the usage of side cutter and location of shank in a press tool	15	-	-	-
PC7. Perform Design and drawing of open die and guide plate press tools .	-	10	-	-
PC8. Demonstrate Design and drawing of progressive type of press tools with cutting and Non-cutting operations.	-	30	-	-
PC9. Perform Design and drawing of compound and combination press tool.	-	30	-	-
PC10. Demonstrate design and drawing of fine blanking tool	-	30	-	-
NOS Total	100	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0128
NOS Name	Manufacture and Try-out Press Tools, Jigs & Fixtures.
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	20
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0120: Design and Manufacture Press Tools for Sheet Metal Components

Description

Able to understand sheet metal processing, Able to design and draw different types of press tools used for manufacturing sheet metal Components.

Scope

The scope covers the following :

- Understand the sheet metal processing by press working operations.
- Understand the different parts of press tool like die set, punch, die block, stripper, pilot, ejector etc.
- Familiar with different types of press tools like progressive tool, compound tool and combination tools.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** • Understand the degrees of freedom, location of job and planes of movements in a jig, fixture.
- PC2.** Understand the difference between jig and fixture and its parts
- PC3.** Learn to Select correct types of jigs and fixtures for a specific operation
- PC4.** Understand tool location with respect to machine in a jig and fixture
- PC5.** Understand mounting and location of a fixture on a machine tool
- PC6.** Learn to Calculate cutting forces acting on jig, fixture
- PC7.** Demonstrate Design and drawing of different types of jigs.
- PC8.** Demonstrate Design and drawing of different types of fixtures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Understand the degrees of freedom, location of job and planes of movements in a jig, fixture.
- KU2.** Understand the difference between jig and fixture and its parts.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn to Calculate cutting forces acting on jig, fixture
- GS2.** Demonstrate Design and drawing of different types of jigs.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	100	-	-
PC1. • Understand the degrees of freedom, location of job and planes of movements in a • jig, fixture.	10	-	-	-
PC2. Understand the difference between jig and fixture and its parts	20	-	-	-
PC3. Learn to Select correct types of jigs and fixtures for a specific operation	20	-	-	-
PC4. Understand tool location with respect to machine in a jig and fixture	15	-	-	-
PC5. Understand mounting and location of a fixture on a machine tool	15	-	-	-
PC6. Learn to Calculate cutting forces acting on jig, fixture	20	-	-	-
PC7. Demonstrate Design and drawing of different types of jigs.	-	50	-	-
PC8. Demonstrate Design and drawing of different types of fixtures.	-	50	-	-
NOS Total	100	100	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0120
NOS Name	Design and Manufacture Press Tools for Sheet Metal Components
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0121: Design and Utilize Jigs & Fixtures for Mass Production

Description

Able to Understand the Functioning of Jig and fixture for mass production. Able to design and draw Jigs & Fixtures.

Scope

The scope covers the following :

- Understand the functioning of jigs and fixtures which are used for making multiple parts of similar accuracy.
- Understand the different types of jigs and its parts used for production purpose.
- Understand the different types of fixtures and its parts used for different operations.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Manufacture of a Progressive tool
- PC2.** Manufacture of a compound tool
- PC3.** Manufacture of a Jig
- PC4.** Manufacture of a Fixture.
- PC5.** Learn to measure tool parts using CMM
- PC6.** Manufacture of Tool/Mould at industry

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Learn to measure tool parts using CMM
- KU2.** Manufacture of a compound tool

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn to measure tool parts using CMM

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	160	80	60
PC1. Manufacture of a Progressive tool	-	40	-	10
PC2. Manufacture of a compound tool	-	40	-	10
PC3. Manufacture of a Jig	-	30	-	10
PC4. Manufacture of a Fixture.	-	30	-	10
PC5. Learn to measure tool parts using CMM	-	20	-	-
PC6. Manufacture of Tool/Mould at industry	-	-	80	20
NOS Total	-	160	80	60

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0121
NOS Name	Design and Utilize Jigs & Fixtures for Mass Production
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0122: Program and Operate Advanced Multi-Axis CNC Machines

Description

Understand the concepts of 5-Axis machining. Able to do the programming and manufacturing of components on Multi-Axis CNC lathe, Milling machine.

Scope

The scope covers the following :

- Understand the concepts of CNC machining for intricate components with GD&T systems.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate Parts of CNC 5-axis machines
- PC2.** Demonstrate Part programming of CNC Machines with Multi-Axis.
- PC3.** Demonstrate Advanced tool holding and job setting methods on High end machines.
- PC4.** Execute part program on Turn-mill.
- PC5.** Execute part program on 5-Axis Machine

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Demonstrate Parts of CNC 5-axis machines
- KU2.** Demonstrate Part programming of CNC Machines with Multi-Axis.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Demonstrate Advanced tool holding and job setting methods on High end machines.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Demonstrate Parts of CNC 5-axis machines	-	5	-	-
PC2. Demonstrate Part programming of CNC Machines with Multi-Axis.	-	20	-	5
PC3. Demonstrate Advanced tool holding and job setting methods on High end machines.	-	15	-	5
PC4. Execute part program on Turn-mill.	-	20	-	5
PC5. Execute part program on 5-Axis Machine	-	20	-	5
NOS Total	-	80	-	20

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0122
NOS Name	Program and Operate Advanced Multi-Axis CNC Machines
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0123: Generate Basic 2D/3D Models and Toolpaths using Master cam

Description

Creating of CNC program by Master cam software and check the tool path Simulation before machining and doing the actual machining of a given profile.

Scope

The scope covers the following :

- Understand the process of creating tool path and NC programme and transferring it to CNC Milling and Lathe machines.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand screen layout and main menu
- PC2.** Learn to Identify the tools for sketching, modelling
- PC3.** Demonstrate Drawing 2D profiles and Modelling.
- PC4.** Learn to Designing with wire frame
- PC5.** Perform Designing and drawing 3D profile.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand screen layout and main menu
- KU2.** Learn to Identify the tools for sketching, modelling

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Perform Designing and drawing 3D profile.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	50	40	-	10
PC1. Understand screen layout and main menu	10	5	-	-
PC2. Learn to Identify the tools for sketching, modelling	10	10	-	5
PC3. Demonstrate Drawing 2D profiles and Modelling.	10	10	-	-
PC4. Learn to Designing with wire frame	10	10	-	5
PC5. Perform Designing and drawing 3D profile.	10	5	-	-
NOS Total	50	40	-	10

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0123
NOS Name	Generate Basic 2D/3D Models and Toolpaths using Master cam
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N6912: Create Basic 2D/3D Models using SolidWorks

Description

Able to draw 2D drawing using Solid works software. Able to Model 3D object using Solid works software.

Scope

The scope covers the following :

- Familiar with different toolbars and functions of Solid works software.
- Understand the Basics of 2D Drawing, Modelling (Surface &Solid) by Solid works software.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate the tools for sketching in Solid works.

PC2. Demonstrate the tools for Basic Modelling in Solid works

PC3. Perform the method of advanced modelling

PC4. Demonstrate the Concept of surface modelling.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Perform the method of advanced modelling

KU2. Demonstrate the tools for sketching in Solid works.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Perform the method of advanced modelling

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Demonstrate the tools for sketching in Solid works.	-	20	-	10
PC2. Demonstrate the tools for Basic Modelling in Solid works	-	20	-	-
PC3. Perform the method of advanced modelling	-	20	-	10
PC4. Demonstrate the Concept of surface modelling.	-	20	-	-
NOS Total	-	80	-	20

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N6912
NOS Name	Create Basic 2D/3D Models using SolidWorks
Sector	Capital Goods
Sub-Sector	
Occupation	Capital Goods, Capital Goods
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0124: Design Complex Press Tools, Jigs & Fixtures using Advanced SolidWorks

Description

Draw different assemblies, Surface modelling and designing of press tools using Solid works software.
Draw different assemblies, Surface modelling and designing of jig/Fixture using Solid works software

Scope

The scope covers the following :

- Understand different modules used in assembly drawing and sheet metal processing.
- Understand drafting with GD&T.
- Practice designing and drawing of different types of Press tools using Solid works.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate the tools for Assembling

PC2. Demonstrate Drafting (2D to 3D)

PC3. Perform Sheet metal module

PC4. Demonstrate Design of press tool with travelling stripper

PC5. Demonstrate Design of progressive tool/compound tool/Jig/Fixture

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Demonstrate the tools for Assembling

KU2. Demonstrate Drafting (2D to 3D)

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Perform Sheet metal module

GS2. Demonstrate the tools for Assembling

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	80	-	20
PC1. Demonstrate the tools for Assembling	-	10	-	5
PC2. Demonstrate Drafting (2D to 3D)	-	10	-	5
PC3. Perform Sheet metal module	-	10	-	10
PC4. Demonstrate Design of press tool with travelling stripper	-	20	-	-
PC5. Demonstrate Design of progressive tool/compound tool/Jig/Fixture	-	30	-	-
NOS Total	-	80	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0124
NOS Name	Design Complex Press Tools, Jigs & Fixtures using Advanced SolidWorks
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0125: Analyze Material Stress, Strain, and Thermal Effects

Description

Understand the effect of stress, strain Understand the effect of temperature on parts/structures.

Scope

The scope covers the following :

- Understand the concept of stress and strain.
- Understand the thermal effect on structures.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Relationship of Stress and strain.
- PC2.** Understand Importance of thermal effect on Stress and Strain.
- PC3.** Learn to Calculate the stress and strain based on load conditions
- PC4.** Analyze the effect of load on struts and columns
- PC5.** Learn to Calculate the deflection due to bending
- PC6.** Learn the effect of torsion on shafts
- PC7.** Understand the effect of torsion on springs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the Relationship of Stress and strain.
- KU2.** Learn to Calculate the stress and strain based on load conditions

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Learn the effect of torsion on shafts

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Understand the Relationship of Stress and strain.	10	-	-	-
PC2. Understand Importance of thermal effect on Stress and Strain.	10	-	-	-
PC3. Learn to Calculate the stress and strain based on load conditions	30	-	-	-
PC4. Analyze the effect of load on struts and columns	10	-	-	-
PC5. Learn to Calculate the deflection due to bending	20	-	-	-
PC6. Learn the effect of torsion on shafts	10	-	-	-
PC7. Understand the effect of torsion on springs	10	-	-	-
NOS Total	100	-	-	-

Logo
Not
Available



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0125
NOS Name	Analyze Material Stress, Strain, and Thermal Effects
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0126: Apply Core Principles of Organizational Management

Description

Familiar with the different functions of management which are required for a successful organization.

Scope

The scope covers the following :

- Understand the functioning of different departments in an organization.
- Understand the concept of financial management

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Understand the Importance of Management and Organization

PC2. Understand the importance of planning, coordinating & Controlling

PC3. Understand the Importance of staffing

PC4. Understand to differentiate between Company and Trade

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Importance of Management and Organization

KU2. Understand the importance of planning, coordinating & Controlling

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Understand the importance of planning, coordinating & Controlling

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Understand the Importance of Management and Organization	25	-	-	-
PC2. Understand the importance of planning, coordinating & Controlling	25	-	-	-
PC3. Understand the Importance of staffing	25	-	-	-
PC4. Understand to differentiate between Company and Trade	25	-	-	-
NOS Total	100	-	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0126
NOS Name	Apply Core Principles of Organizational Management
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Bench Fitter
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

TNSD/CSC/N0127: Perform Tool & Die Cost Estimation and Economic Analysis

Description

Find out the approximate/actual cost of components, Mould and dies by costing procedures. Find out the approximate/actual cost Mould and dies by costing procedures

Scope

The scope covers the following :

- Understand the difference between estimating and costing.
- Understand the different components which constitutes the cost.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Importance of estimation and costing
- PC2.** Understand the difference between estimation and costing
- PC3.** Learn the Components of the cost and selling price of an item
- PC4.** Learn to calculate Basic machine hour rate
- PC5.** Understand the terminology used in business transactions and awareness of accounting.
- PC6.** Learn to calculate cost of Sheet metal and molded components
- PC7.** Estimate the cost of press tools, moulds , dies , jigs and fixtures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand the Importance of estimation and costing
- KU2.** Understand the difference between estimation and costing
- KU3.** Learn to calculate Basic machine hour rate

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Estimate the cost of press tools, moulds, dies, jigs and fixtures
- GS2.** Understand the terminology used in business transactions and awareness of accounting.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	100	-	-	-
PC1. Understand the Importance of estimation and costing	10	-	-	-
PC2. Understand the difference between estimation and costing	10	-	-	-
PC3. Learn the Components of the cost and selling price of an item	10	-	-	-
PC4. Learn to calculate Basic machine hour rate	20	-	-	-
PC5. Understand the terminology used in business transactions and awareness of accounting.	20	-	-	-
PC6. Learn to calculate cost of Sheet metal and molded components	20	-	-	-
PC7. Estimate the cost of press tools, moulds , dies , jigs and fixtures	10	-	-	-
NOS Total	100	-	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	TNSD/CSC/N0127
NOS Name	Perform Tool & Die Cost Estimation and Economic Analysis
Sector	Capital Goods
Sub-Sector	
Occupation	Bench Fitter, Capital Goods, Bench Fitter
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Assessment System Overview:

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping TNSDC
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- TNSDC monitors the assessment process & records

2. Testing Environment:

- Check the Assessment location, date and time
- If the batch size is more than 30, then there should be 2 Assessors.
- Check that the allotted time to the candidates to complete Theory & Practical Assessment is correct.

3. Assessment Quality Assurance levels/Framework:

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- Question bank is created by the Subject Matter Experts (SME) are verified by the other SME
- Questions are mapped to the specified assessment criteria
- Assessor must be ToA certified & trainer must be ToT Certified

4. Types of evidence or evidence-gathering protocol:

- Time-stamped & geotagged reporting of the assessor from assessment location
- Centre photographs with signboards and scheme specific branding

5. Method of verification or validation:

- Surprise visit to the assessment location

6. Method for assessment documentation, archiving, and access

- Hard copies of the documents are stored

On the Job:

- Each module will be assessed separately.
- The candidate must score 60% in each module to successfully complete the OJT.
- Tools of Assessment that will be used for assessing whether the candidate is having desired skills and etiquette of dealing with customers, understanding needs & requirements, assessing the customer and perform Soft Skills effectively:
- Videos of Trainees during OJT
- Assessment of each Module will ensure that the candidate is able to:
- Effective engagement with the customers
- Understand the working of various tools and equipment

Minimum Aggregate Passing % at QP Level : 60

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
TNSD/CSC/N0128.Manufacture and Try-out Press Tools, Jigs & Fixtures.	100	100	0	0	200	20
TNSD/CSC/N0120.Design and Manufacture Press Tools for Sheet Metal Components	100	100	0	0	200	10
TNSD/CSC/N0121.Design and Utilize Jigs & Fixtures for Mass Production	0	160	80	60	300	10
TNSD/CSC/N0122.Program and Operate Advanced Multi-Axis CNC Machines	0	80	0	20	100	10
TNSD/CSC/N0123.Generate Basic 2D/3D Models and Toolpaths using Master cam	50	40	0	10	100	10
TNSD/CSC/N6912.Create Basic 2D/3D Models using SolidWorks	0	80	0	20	100	10
TNSD/CSC/N0124.Design Complex Press Tools, Jigs & Fixtures using Advanced SolidWorks	0	80	0	20	100	10
TNSD/CSC/N0125.Analyze Material Stress, Strain, and Thermal Effects	100	0	0	0	100	5
TNSD/CSC/N0126.Apply Core Principles of Organizational Management	100	0	0	0	100	5
TNSD/CSC/N0127.Perform Tool & Die Cost Estimation and Economic Analysis	100	0	0	0	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	570	670	80	130	1450	100

Logo
Not
Available



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.